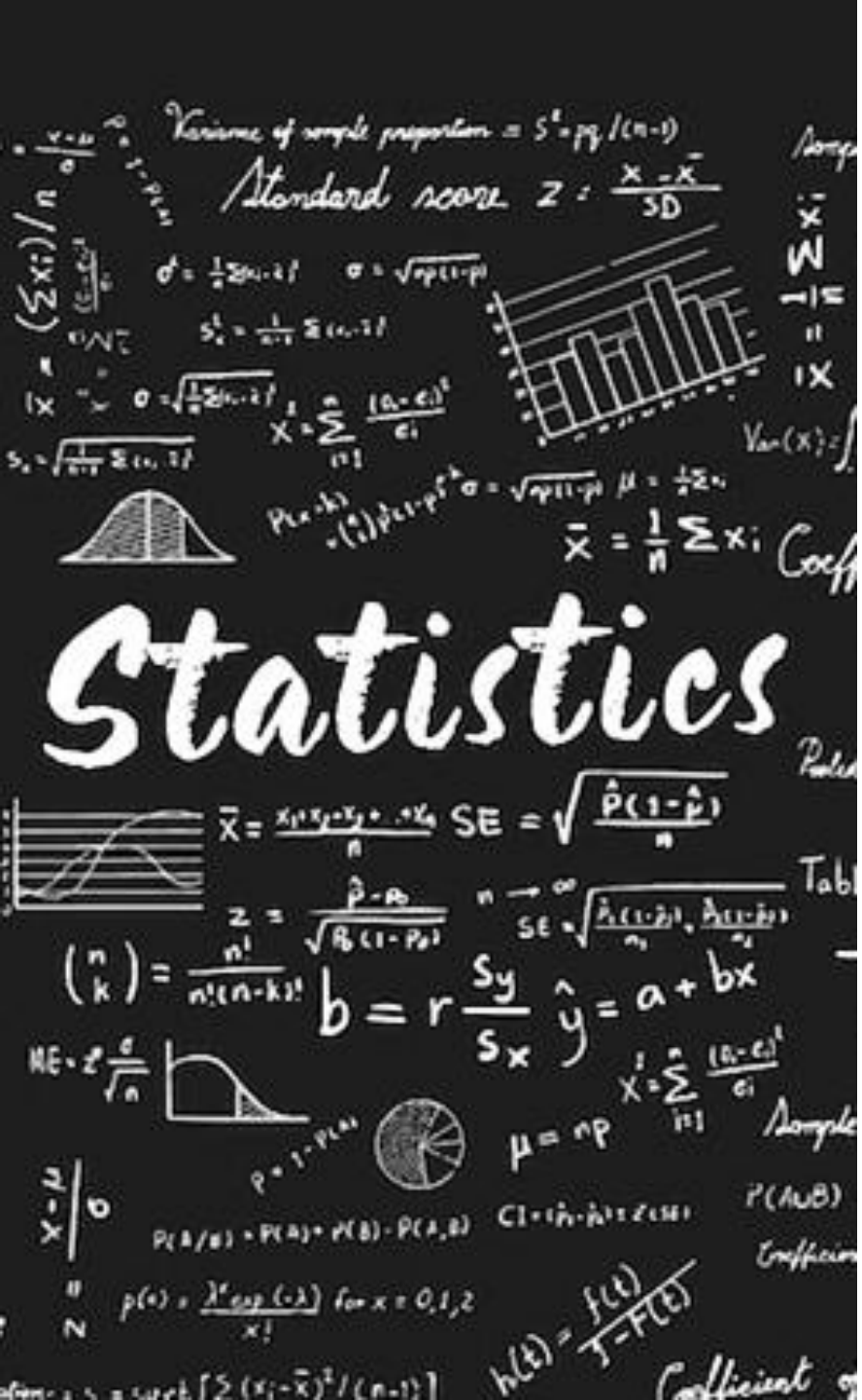
Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
Asad Ullah
- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)



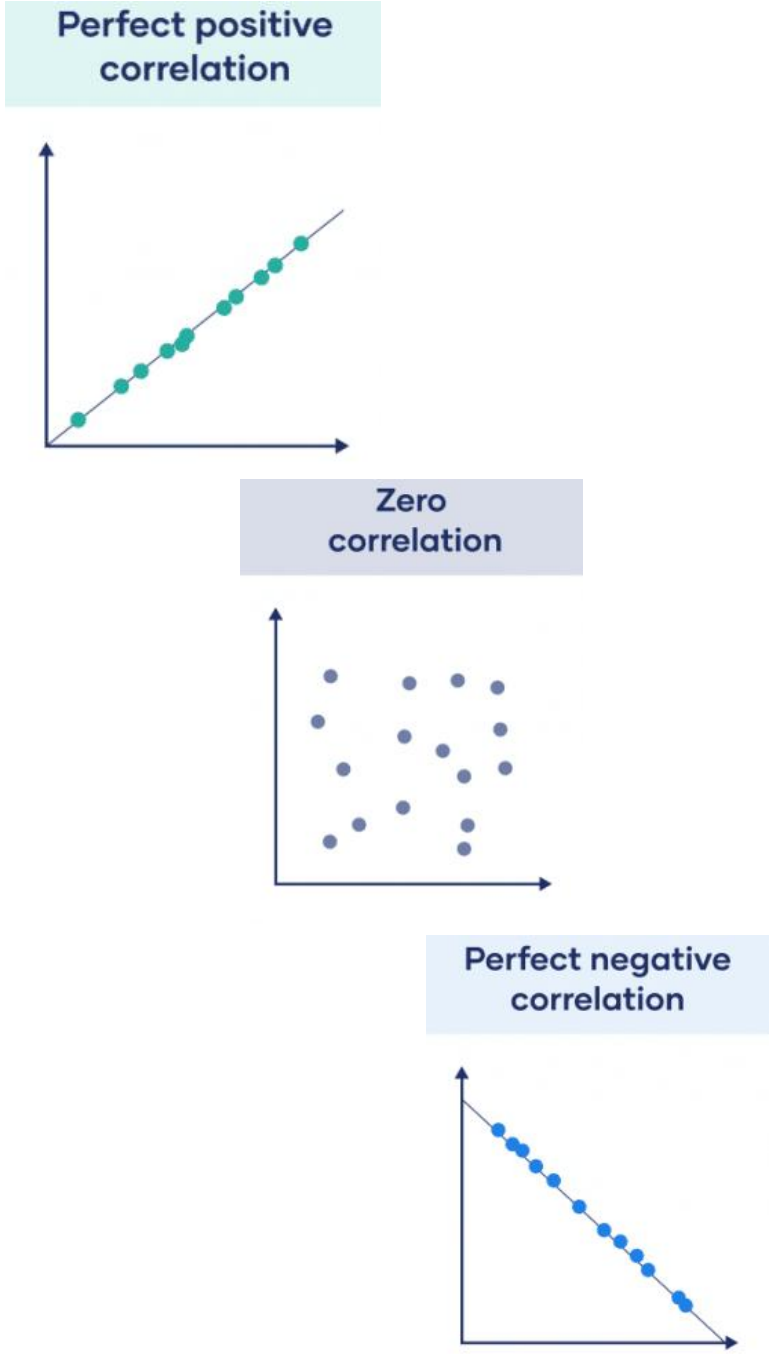
Day 01 – Descriptive Statistics
(Correlation)

Objectives:

- What is Correlation?
- Correlation Coefficient
- Univariate, Bivariate, and Multivariate Plots
- Introduction to Probability

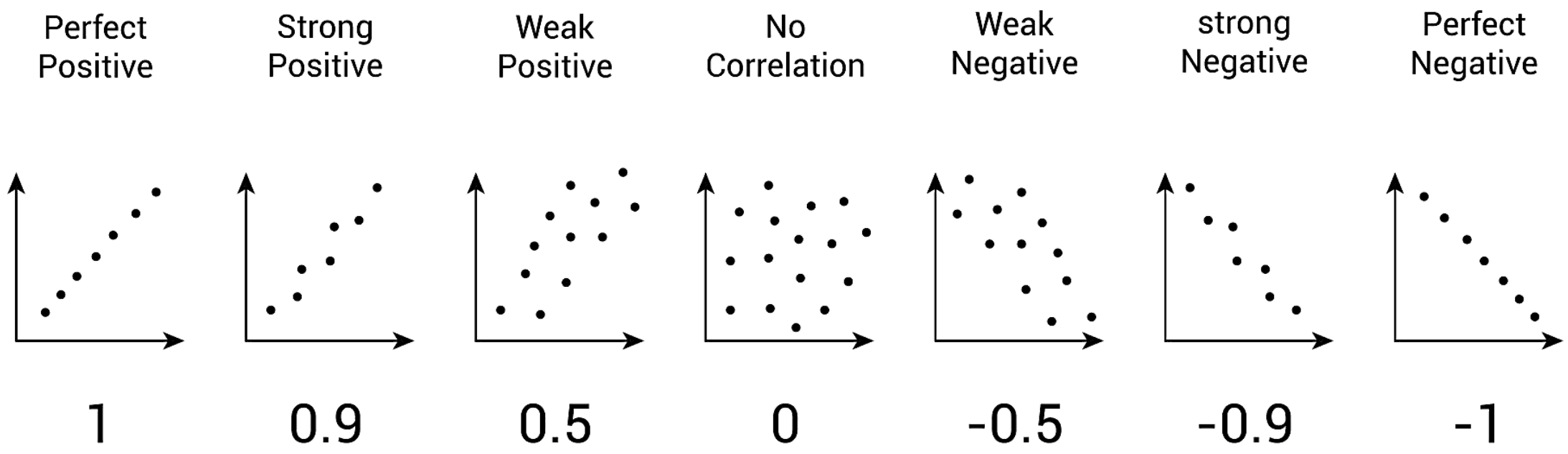
Correlation

- A statistical measure that describes the strength and direction of the relationship between two variables, showing how they tend to change together.
- It's quantified by a correlation coefficient.
- Correlation may be positive, zero, or negative.



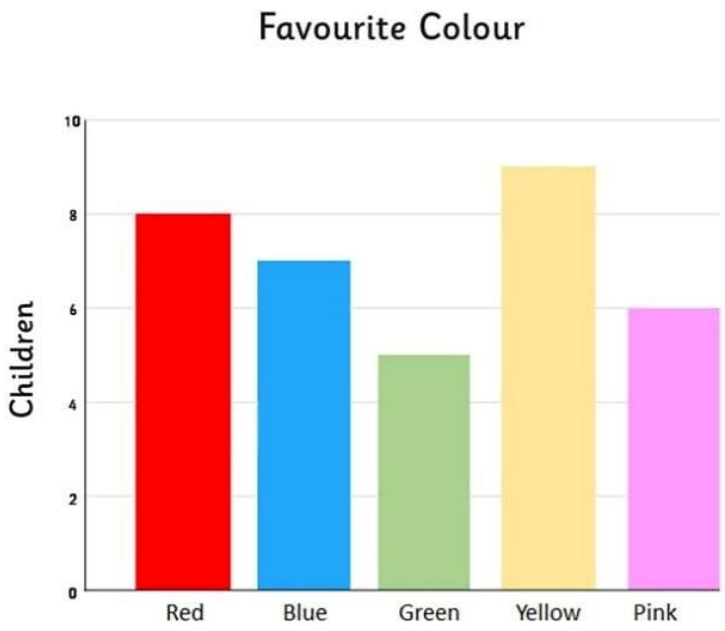
Correlation Coefficient

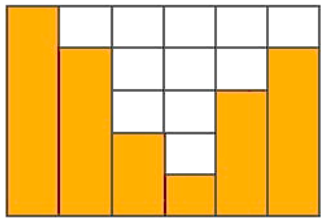
A correlation coefficient is a number between -1 and 1 that tells you the strength and direction of a relationship between variables.



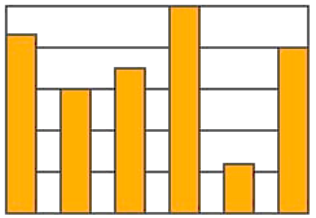
Univariate Data

- It consists of observations for a single characteristic or variable, used to describe and summarize its distribution, central tendency, and variability.
- Data can be displayed using frequency polygon, histograms, stem and leaf plots, box and whisker plots, or pie charts.

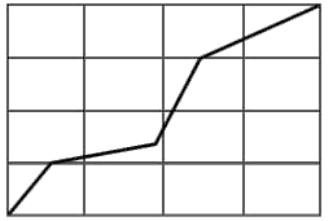




Histogram



Bar Chart



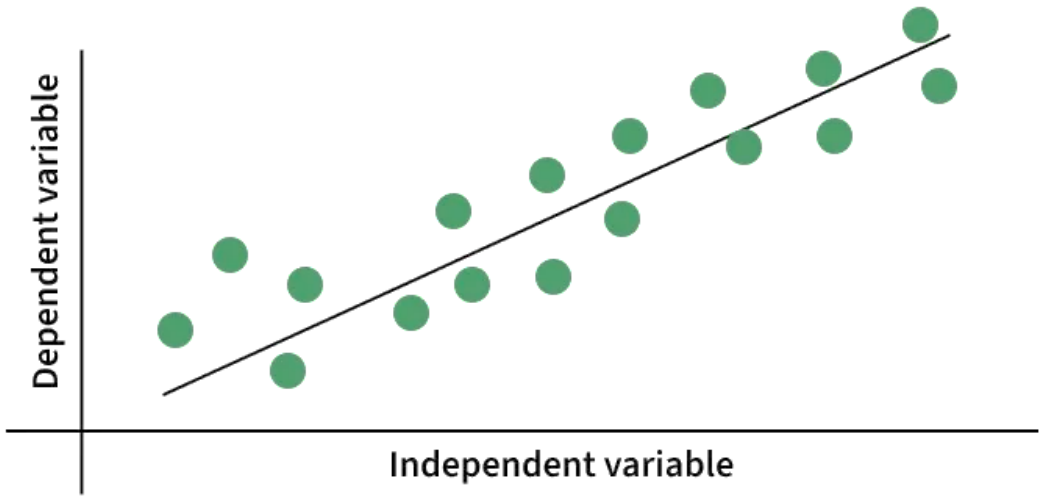
Frequency polygon

Pie Chart

Bivariate Data

- It consists of two variables where each data point has a value for both, used to analyze the relationship between them.
- Often analyzed using a scatterplot, where one variable is plotted on the x-axis and the other on the y-axis.
- Each data point represents two pieces of information.

Scatterplot



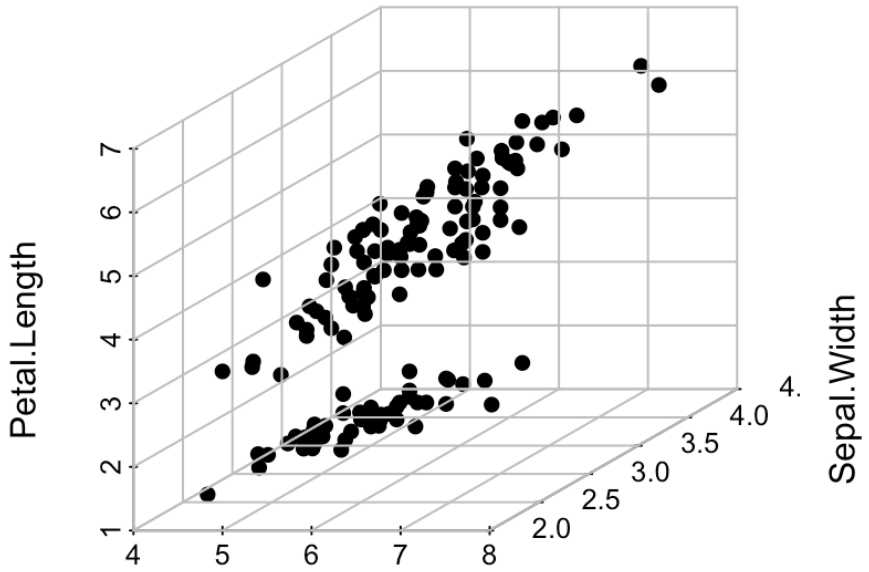
Years of Schooling	Annual Income
12	$36,000
11	$32,000
16	$58,000
16	$65,000
16	$76,000
18	$89,000
17	$45,000
20	$84,000
17	$125,000
...	...

Multivariate Data

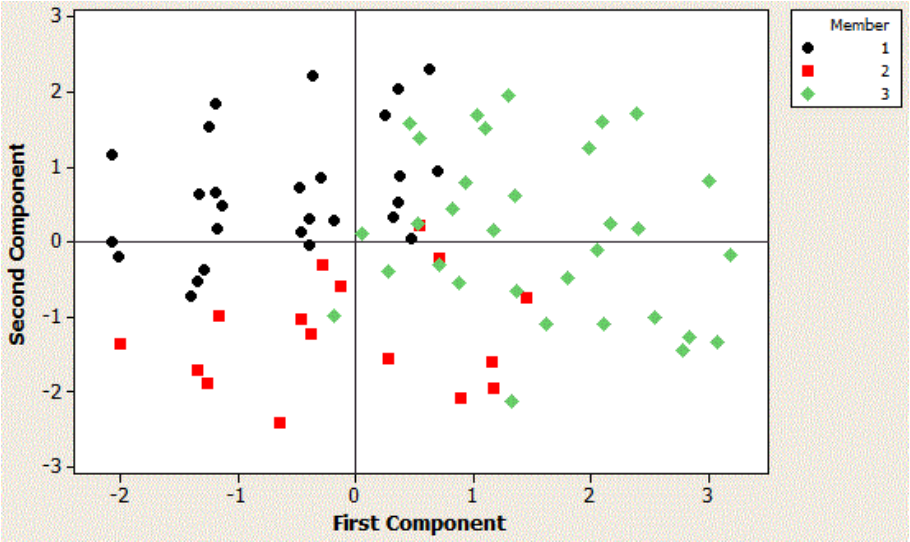
- It consists of observations for two or more variables for each unit in a dataset, such as a person's age, weight, and blood pressure.

Household ID	Household Size	Annual Income	Number of Pets
1	2	$37,000	0
2	4	$49,000	0
3	4	$58,000	1
4	1	$68,000	3
5	3	$61,000	2
6	5	$64,000	2
7	6	$79,000	1
8	4	$89,000	1
9	7	$104,000	1
10	2	$95,000	0

Scatterplot

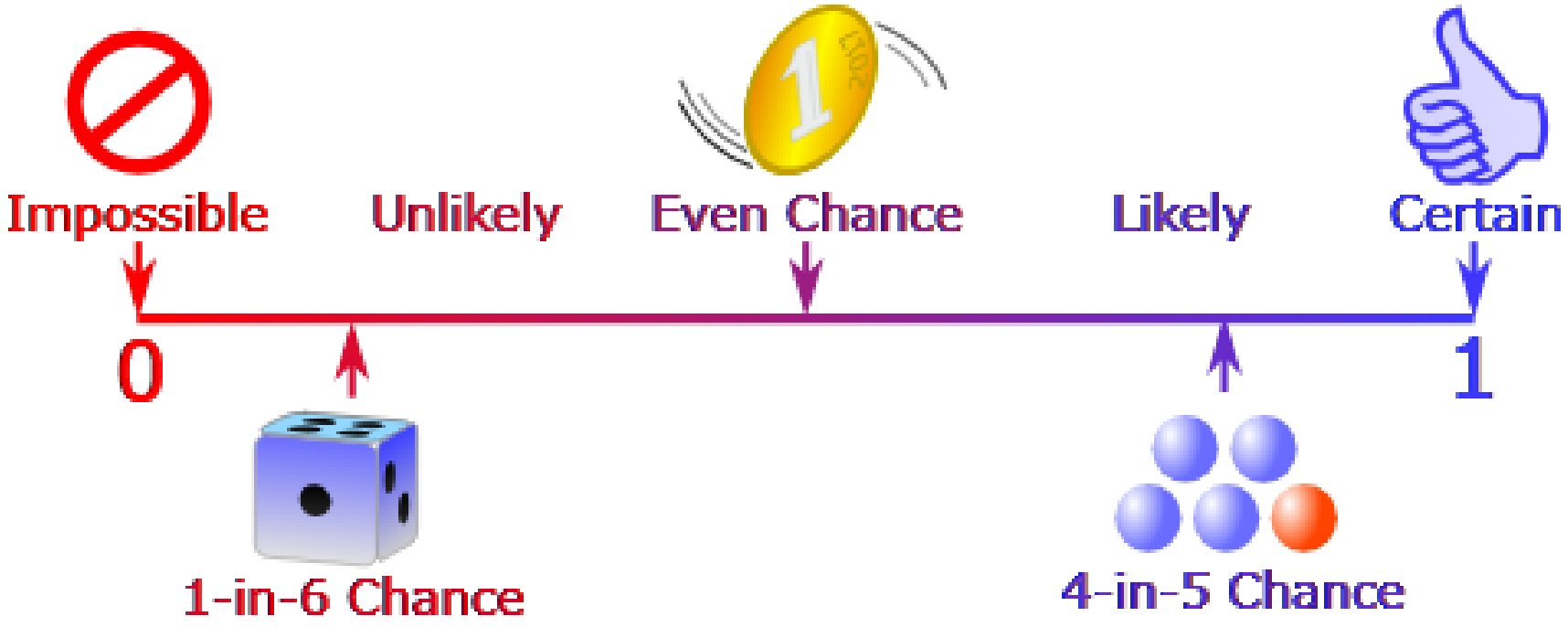


Sepal.Length



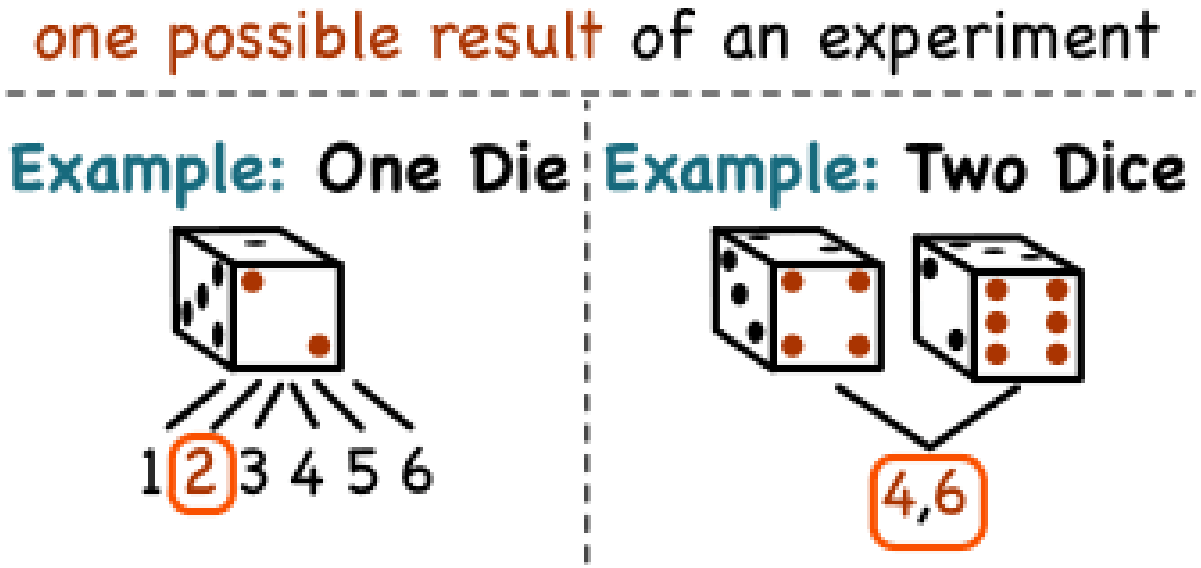
Probability

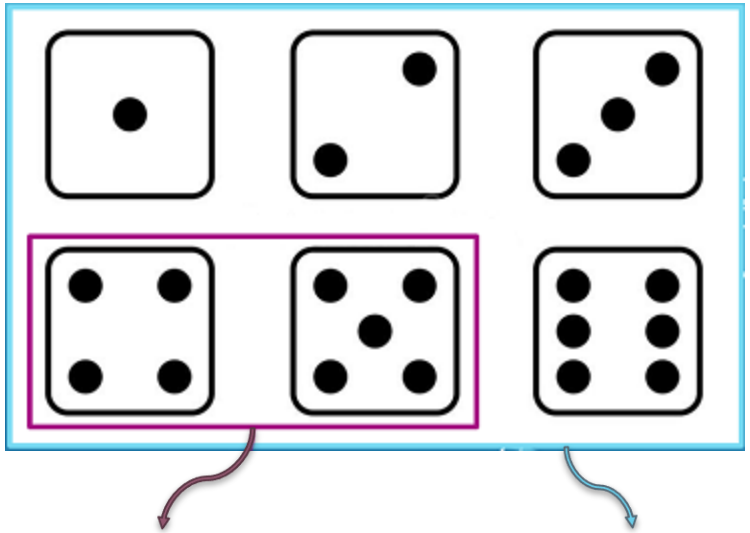
- Probability is the measure of how likely an event is to occur.
- It is expressed as a number between 0 and 1, or 0% and 100%.
- Formula for probability of single event:
  P(E) = number of ways event can happen / number of possible events



Sample Space and Events

- **Sample space (S)** is the complete set of all possible outcomes for a random experiment.
- An **event** is a specific subset of the sample space, representing one or more of those possible outcomes





Probability Distributions

- **Sample space (S)** is the complete set of all possible outcomes for a random experiment.
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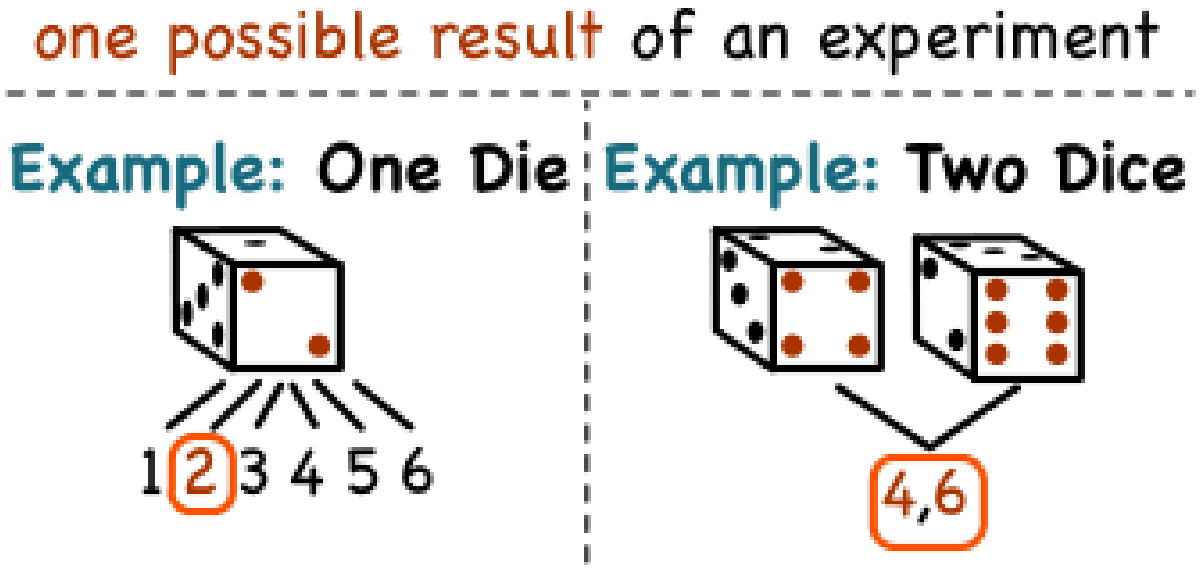

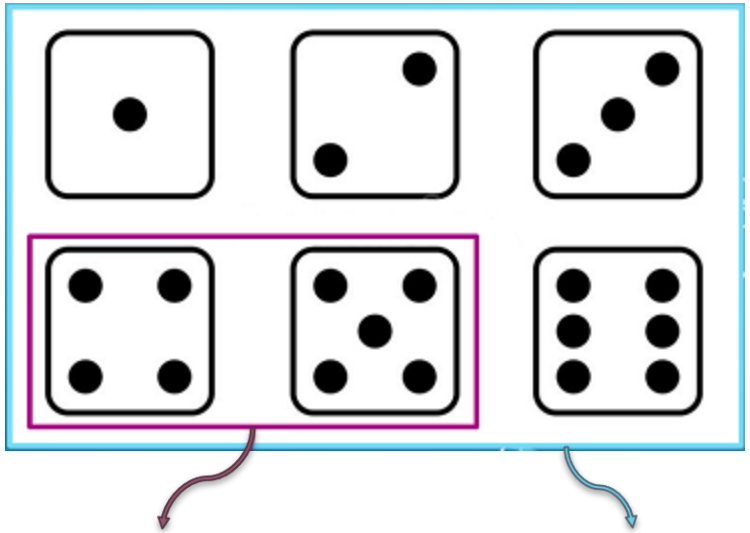

Thank You